# Huijun Jin

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*Ph.D. expected February 2027. Available for a postdoctoral position from March 2027.*

## Research Interests

Database systems and vector data management: graph-based approximate nearest neighbor search (HNSW), index maintenance under updates, robustness to out-of-distribution queries, hardware-conscious index layouts, retrieval-augmented generation, and machine learning for database tuning.

## Education

**Ph.D. in Computer Science** Mar. 2024 – Feb. 2027 (expected)
Yonsei University, Seoul, Korea
Dissertation: *Optimizing Efficiency, Updatability, and Robustness of HNSW Indexes for High-Performance Vector Search*
Advisor: Prof. Sanghyun Park. Preliminary defense passed; final defense planned for Nov.–Dec. 2026.

**M.S. in Computer Science** Mar. 2020 – Feb. 2022
Yonsei University, Seoul, Korea. Advisor: Prof. Sanghyun Park

**B.Eng. in Computer Science and Technology** Sep. 2014 – Jun. 2018
Yanbian University, Yanji, China

## Publications

**Selected publications (first author)**

[1] **Huijun Jin**, Sanghyun Park. CAMP-HNSW: Hardware-Topological Co-Design for OOD-Robust ANNS via Cache Alignment and Targeted Rewiring. *Proceedings of the VLDB Endowment (PVLDB)*, 20(2):438–451, 2026. To be presented at VLDB 2027. [code]

[2] **Huijun Jin**, Jieun Lee, Shengmin Piao, Sangmin Seo, Sanghyun Park. PRO-HNSW: Proactive Repair and Optimization for High-Performance Dynamic HNSW Indexes. In *Proc. IEEE International Conference on Data Engineering (ICDE)*, 2026. [code]

[3] **Huijun Jin**, Jieun Lee, Shengmin Piao, Sangmin Seo, Sein Kwon, Sanghyun Park. Efficient Approximate Nearest Neighbor Search via Data-Adaptive Parameter Adjustment in Hierarchical Navigable Small Graphs. In *Proc. Design, Automation & Test in Europe Conference (DATE)*, 2025. [code]

**Other peer-reviewed publications**

[4] Sein Kwon, Youngwan Jo, Seungyeon Choi, Jieun Lee, **Huijun Jin**, Sanghyun Park. LatentTune: Efficient Tuning of High Dimensional Database Parameters via Latent Representation Learning. In *Proc. IEEE International Conference on High Performance Computing, Data, and Analytics (HiPC)*, 2025.

[5] Jieun Lee, Sangmin Seo, Chanho Yeom, **Huijun Jin**, Sein Kwon, Sanghyun Park. An Efficient and Noise-Robust Framework for High-Dimensional Tuning in Big Data Analytics. *Engineering Applications of Artificial Intelligence*, 158:111332, 2025.

[6] **Huijun Jin**, Jieun Lee, Sanghyun Park. RTune: A RocksDB Tuning System with Deep Genetic Algorithm. In *Proc. Genetic and Evolutionary Computation Conference (GECCO)*, 2022.

[7] **Huijun Jin**, Won Gi Choi, Jonghwan Choi, Hanseung Sung, Sanghyun Park. Improvement of RocksDB Performance via Large-Scale Parameter Analysis and Optimization. *Journal of Information Processing Systems*, 18(3):374–388, 2022.

**Publications in Korean venues**

Seven papers in Korean venues (Communications of the KIISE, KIISE Transactions on Computing Practices, KCC, KIPS), including a first-author survey on retrieval-augmented generation and vector databases in a special issue of Communications of the KIISE (2024). Full list at huijunjin.github.io/publications.

## Research Experience

**Data Engineering Lab, Yonsei University** (Advisor: Prof. Sanghyun Park)

- Ph.D. researcher — Mar. 2024 – present
- Researcher (post-master's) — Mar. 2022 – Aug. 2022
- M.S. researcher — Mar. 2020 – Feb. 2022
- Participant, S/W Starlab program (IITP, Korea): software development and database testing

## Conference Presentations

- CAMP-HNSW — VLDB 2027, Athens, Greece (scheduled, Aug. 2027)
- PRO-HNSW — ICDE 2026, Montréal, Canada
- Data-adaptive HNSW search — DATE 2025, Lyon, France
- RTune — GECCO 2022, Boston, USA

## Teaching

Teaching Assistant, *Database Systems*, Yonsei University — Spring 2020

## Skills and Languages

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|---|---|
| **Programming** | C++ (index engines), Python (experiment pipelines and analysis) |
| **Systems** | Graph-based vector indexes (HNSW), RocksDB, Redis |
| **Languages** | Korean (native), Chinese (native), English (professional working proficiency) |

## References


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